Neelan Chung

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Skills

Software & Programming: C/C++, Python, Java, MATLAB, Git, Unix, STM32CubeMX/IDE
Electrical: Circuit Design (Altium, LTspice), Hardware Communication Protocols (CAN, I2C, UART, SPI, SWD)
Hardware & Design: CAD (SolidWorks, BricsCAD), Design Review (Navisworks, Intergraph Smart Review)

Education

University of British Columbia — Expected May 2030
B.A.Sc. Integrated Engineering - GPA: 4.33 — *Vancouver, BC*

Experience

Mechanical & Materials Engineering Intern — Aug 2026 – Present
Fluor Canada — *Vancouver, BC*

- 3D modelled **mining infrastructure and bulk-material flows** in **SolidWorks** for large-scale mining projects
- Managed **vendor package documentation**, routed documents for signatures and project management review, and developed a VBA tool to rename, organize, and designate vendor files
- Assisted with **gyratory crusher and jib crane equipment packages**, supporting package engineers and the material handling team with technical coordination and documentation

Telemetry Team Lead & Firmware Member — Sept 2025 – Present
UBC ThunderBikes — *Vancouver, BC*

- Developed **embedded systems** using **STM32** microcontrollers, STM32CubeMX, and the Makefile build tool
- Currently designing and writing the firmware for a **custom telemetry PCB** to record bike metrics
- Wrote firmware for a **CAN-based charger monitoring module** to display battery charging values over time
- Assisted with CAD modelling and prototyping for electronic enclosures and charging-port housings
- Collaborated with firmware, electrical, and mechanical sub-teams on system integration and prototyping

Projects

Smart Navigation Goggle Module — *2nd-Year IGEN Capstone* — Jan 2026 – Apr 2026

- Developed object-oriented embedded firmware in **C++** for an **ESP32-S3** wearable Heads-Up Display (HUD) using custom consumer methods, display driver classes, and handling structures with the Arduino framework and IDE
- Implemented **BLE** communication architecture using **GATT** services and characteristics for real-time transmission of navigation and performance data from an Android app that took navigation data from the Google Maps API
- Built data-processing classes to render live pace, distance, timing, and directional metrics on an **OLED** display
- Developed firmware using **ESP32 BLE APIs** and validated stable wireless communication and timed updates
- Contributed to a multidisciplinary project spanning embedded systems, optics, CAD, and wearable hardware prototyping

File Renaming and Organization Program — *Fluor Mining & Metals* — Sept 2026

- Created a file-renaming program in VBA to rename vendor package files using vendor-specified metadata.
- Built a UserForm-based interface that guided users through setting up project folders and selecting metadata rows and columns at runtime.
- Used VBA UserForms, file system objects, dictionaries, and error handling methods to create a renaming manifest, allowing the user to input missing information, customize file names, and correct potential renaming errors
- Used a modular, segmented approach to developing the program, carefully tracking and managing object scope

CAN-Controller Charger Voltage Display — *UBC ThunderBikes* — Sept 2025

- Used STM32CubeMX to program an **STM32 Blue Pill** to receive battery charging data, including cell voltage and current, from the bike's **CAN bus** and display it on an OLED screen in a metric-over-time format
- Collaborated with the BMS team on firmware development and designed a custom housing using **SolidWorks**

Mining Infrastructure CAD — *Fluor Mining & Metals* — Aug 2026 - Sept 2026

- Designed and modeled dump-pocket and surge-pocket infrastructure in **SolidWorks** for Fluor's El Abra Mine Project, and used surface modeling, feature intersections, and constraints to estimate material volume and fill line locations
- Developed flexible, equation-driven bulk-material fill geometry with multiple configurations to preserve design intent